

(13) – Magazine Management System

Supervisor TA: Akram Tarek

Date:27/3/2025

Name	section	Department	ID
اميرة مصطفى هارون عبدالوهاب	1	CS	2022170077
سهام سيد حسن حسين	3	IS	2022170193
فاطمة محمود عبدالقادر القصبي	4	CS	2022170304
اسراء سعيد شعبان حجاج	1	CS	2022170056
منة الله علاء احمد محمد	5	CS	2022170627
ليلى محمد ابوالفتوح السيد	4	CS	2022170326
سعاد سعيد ابراهيم محمد	3	SC	2022170185

1. Introduction

The Magazine Management System is a digital platform designed to simplify content management, enhance reader engagement, and support administrative control over digital articles.

It supports two main roles: Readers and Admins, each with specific functionalities.

Readers can browse articles, react, comment, follow their favorite sections, and receive notifications for the latest articles in it.

Admins can manage articles, moderate reader interactions, and monitor reader accounts.

The system integrates with a database, ensuring real-time updates and effective integration between Reader interface and Admin Panel. This integration allows for efficient content monitoring and management, minimizing manual errors and maintaining platform integrity.

By optimizing content handling, enhancing reader interaction, and ensuring effective moderation, the system aligns with the organization's goals of improving content quality, increasing reader engagement, and being a trusted platform.

This approach supports the organization's strategic goal of building a reliable and engaging digital platform for its audience.

2. User Requirements

○ Readers Requirements

■ Account Management:

- Readers can register and manage their accounts.
- Readers can log in and out securely.
- Readers expect secure authentication and a reliable password recovery process.

■ Content Browsing and Interaction:

- Readers can read and search for articles using filters (title, publisher name, and date).
- Readers can react (like/dislike) and comment on articles.
- Readers can save their favorite articles for quick access.

■ Notifications:

- Readers can follow specific sections.
- Readers receive notifications for new articles added to their followed sections.

○ Admin Requirements

■ Account Management

- Admins can log in, register, edit their profile, delete their profile, and log out, similar to readers.
- Admins must enter a secret admin code during registration to verify their identity and prevent unauthorized admin accounts.
- **Reader Management:**
 - Admins can view and delete reader accounts.
 - Admins can moderate inappropriate content, comments and reactions.
- **Content Management:**
 - Admins can create, edit, and delete articles and sections.
 - Admins can notify readers about new content published in the sections they follow.

3. Functional Requirements

a. Account Access and Credential Management

- **Description/Action:**
Enable Users (both readers and admins) to register, securely log in, and recover their accounts through a password reset process.
- **Requirements/Inputs:**
 - **Registration:** Provide a username, password, email, phone number, security question and answer (for password recovery), and an optional profile picture.
 - **Admin Registration:** Admins must provide a valid secret code along with regular registration details.
 - **Login:** Enter a valid username and password.
 - **Password Reset:** Provide the registered email, answer the security question, and set a new password that meets the defined criteria.
- **Source:**
Accessed through the homepage's registration, login, and password reset forms.
- **Pre-condition:**
 - **Registration:** Ensure the provided email contains "@" and ".com" for validity.
 - **Password Criteria:**
 - Minimum 8 characters.
 - At least one uppercase letter.
 - At least one lowercase letter.
 - At least one special character (!@#\$%^&*).
 - At least one number (0-9).
 - **Login:** Ensure the Reader has a registered account with valid credentials.
 - **Admin Registration:** Validate the secret admin code.
 - **Password Reset:** Verify the registered email and the correct answer to the security question.

■ **Post-condition:**

- Create a new Reader account upon successful registration and redirect to the login page.
- Grant full access to the personalized Reader dashboard upon successful login.
- Display an error message if login fails, with options to retry or reset the password.
- Update the Reader's credentials after a successful password reset and redirect to the login page.

■ **Output:**

- Display a confirmation message upon successful registration and password reset.
- Provide access to the personalized Reader dashboard and profile.
- Show an error notification if login fails.

b. Article Browsing, Search, and Interaction

■ **Description/Action:**

Allow Readers to explore and search for articles using various filters, interact through reactions and comments, save articles to favorites, and receive notifications for followed sections.

■ **Requirements/Inputs:**

- **Search:** Use keywords like title, author, or date.
- **Interaction:** Select reactions (like), add comment text, and save to favorites.
- **Follow/Unfollow Section:** To receive notifications.

■ **Source:**

Search bar on the homepage, interaction options under articles, and follow buttons on section pages.

■ **Pre-condition:**

- Ensure the Reader is logged in to view full articles, react, comment, save to favorites, and receive notifications.
- Verify the Reader follows a section to receive notifications for new content.

■ **Post-condition:**

- Display search results matching the criteria.
- Update reaction counts, show comments, and save articles to the Reader's favorites.
- Show notifications in the Reader profile, sorted by the latest first.

■ **Output:**

- Display a list of articles matching search criteria with relevant details (title, author, and publication date).

- Enhance Reader engagement through interactions like reactions, comments, and saved favorites.
- Timely notifications for new content.

c. User Profile Management

■ **Description/Action:**

Allow admins to view, update and delete their accounts permanently, readers can do the same, in addition to access their favorite list and notification history.

■ **Requirements/Inputs:**

- **Update Information:** Provide updated profile details (username, password, email, phone number, profile picture).
- **Delete Account:** Enter password and the answer to the security question, and for admin enter his secret code.

■ **Source:**

Accessible through the reader profile page after a successful login.

■ **Pre-condition:**

Ensure the reader is logged in to view, edit, or delete their profile.

■ **Post-condition:**

- Save updated profile information and display an updated favorite list and notifications.
- Permanently delete the account and associated data (comments, reactions, favorites, and profile details) if requested, and log the reader out after account deletion.

■ **Output:**

- Display a personalized profile with updated information.
- Provide a confirmation message upon successful account deletion.

d. Admin Account Management and Content Moderation

■ **Description/Action:**

Enable admins to manage reader accounts, moderate content, delete inappropriate comments, and ensure a respectful community.

■ **Requirements/Inputs:**

- **Admin Login:** Enter valid username, password, and admin code during registration.

■ **Source:**

Admin panel accessible upon successful login, with moderation tools.

- **Pre-condition:**
Authenticate the admin with valid credentials and the admin code.
- **Post-condition:**
 - Allow admins to view and delete reader accounts.
 - Remove inappropriate or offensive comments.
- **Output:**
 - Maintain a controlled and respectful platform environment.
 - Ensure secure, regulated access for admins.

e. Article and Section Management

- **Description/Action:**
Allow admins to create, edit, delete articles and sections, and notify readers following sections when new content is published.
- **Requirements/Inputs:**
 - **Article Management:** Provide title, content, and images.
 - **Section Management:** Provide section name and description.
 - **Editing and Deleting:** Specify the target article or section.
 - **Notification Management:** Select the section and target reader group for notifications.
- **Source:**
Admin panel with content creation and notification tools.
- **Pre-condition:**
Require admin authentication to access content management features.
- **Post-condition:**
Display new or updated content to all readers and notify followers.
- **Output:**
Provide dynamic, updated content across the platform.

4. Non-Functional Requirements

- **Extensibility:**
Support adding new sections and articles without major changes.
- **Security:**
 - Implement a strong password policy:
 - Minimum 8 characters, including uppercase, lowercase, special character, and number.

- Use a secure password reset process via security questions.
- Validate email format and ensure unique usernames and emails.
- Admin registration requires a secret code to verify system authorization before account creation. This prevents unauthorized users from registering as admins.

- **Access Control:**

Implement Role-Based Access Control (RBAC) for appropriate permissions.

((RBAC) ensures that only authorized admins, verified by a secret code, can access admin functionalities.)

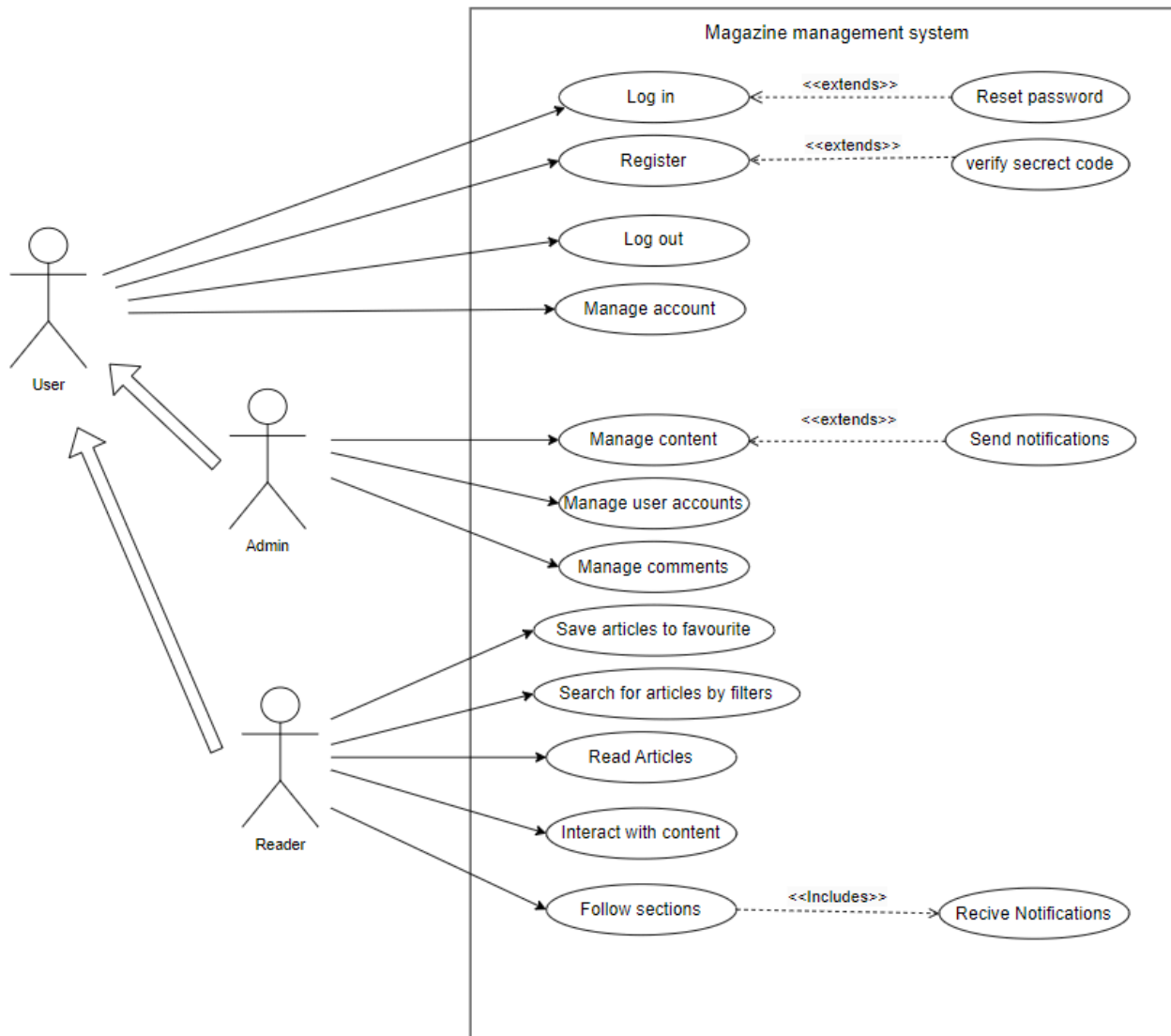
- **Usability:**

- Design a user-friendly interface for navigating articles, sections, and profiles.
- Ensure quick access to frequently used features like the search bar and seamless following of preferred sections.

- **Privacy & Data Management:**

Enable users to permanently delete their accounts and data.

Use case Diagram



Sequence Diagram

